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Case 34-2025: A 57-Year-Old Woman with Visual Disturbances and Right-Arm Shaking

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PRESENTATION OF CASE

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CME



Dr. Ashwin Amurthur (Neurology): A 57-year-old woman was admitted to this hospital because of persistent visual disturbances involving the right visual field, episodes of shaking of the right arm and leg, and hyperintensities on T1-weighted magnetic resonance imaging (MRI) of the head.

The patient had been in her usual state of health until 2 years before the current presentation, when frontotemporal and cervico-occipital headaches developed. The headaches occurred every 1 to 2 months, were associated with photophobia, and typically lasted for several hours to 1 day.

Nineteen months before the current presentation, the patient was evaluated by her primary care physician. The physical examination was normal. A diagnosis of migraines was considered. MRI of the head was performed, and T2-weighted fluid-attenuated inversion recovery (FLAIR) sequences revealed signal abnormalities in the subcortical and periventricular white matter, which were considered to be nonspecific. The patient was evaluated by a neurology consultant, who agreed with the diagnosis of migraines and recommended the use of intranasal zolmitriptan.

Six months before the current presentation, a severe headache developed suddenly while the patient was visiting a family member. She was evaluated at another hospital. Confusion was noted, along with generalized tonic-clonic seizure activity.

Dr. Shenghua Zhu: Computed tomography (CT) of the head revealed hyperdensity concentrated in the sulci of the left parietal lobe, a finding that was thought to be indicative of small-volume convexal subarachnoid hemorrhage (Fig. 1A). CT angiography and venography of the head and neck showed asymmetric prominence of the left distal branches of the anterior cerebral artery that extended to the left posterior parietal region, where the presumed subarachnoid hemorrhage was located, as well as mild prominence of the cortical venous structures in that region — findings consistent with a pial arteriovenous fistula (Fig. 1B, 1C, and 1D). MRI of the head was performed 10 hours after the initial presentation. T1-weighted sequences showed intrinsic hyperintensities that were most prominent along the left parietal lobe,

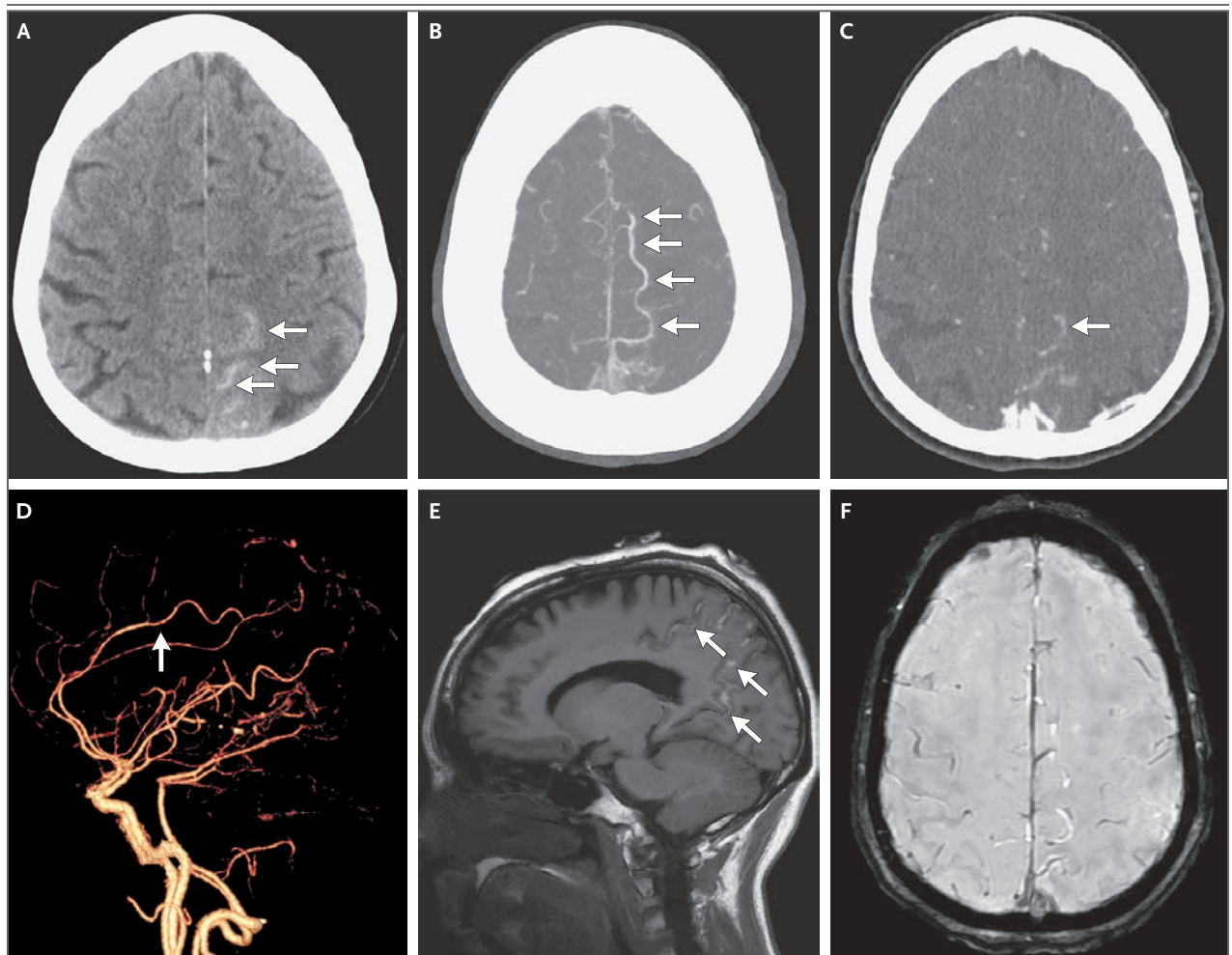


Figure 1. Imaging Studies of the Head, 6 Months before the Current Presentation.

An axial CT image (Panel A) shows hyperdensity concentrated in the sulci of the left parietal lobe (arrows). Axial images from CT angiography and venography (Panels B and C, respectively) show asymmetric prominence of the left distal branches of the anterior cerebral artery (Panel B, arrows) and mild prominence of the cortical veins (Panel C, arrow) in the paracentral parietal lobe. A three-dimensional reformatting image from CT angiography (Panel D) also shows asymmetric prominence of the left anterior cerebral artery (arrow). A sagittal T1-weighted MRI (Panel E) shows intrinsic hyperintensities in the sulci of the left parietal lobe (arrows). An axial susceptibility-weighted angiographic MRI (Panel F) shows no corresponding susceptibility signal in this region.

a finding consistent with the presence of blood products related to subarachnoid hemorrhage (Fig. 1E and 1F).

Dr. Amurthur: Cerebral angiography reportedly revealed a fistula in the left middle meningeal artery that was connected to the superior sagittal sinus. An embolic agent was injected into the fistula, and follow-up angiography showed successful complete occlusion of the fistula. After the procedure, the patient continued to have headaches, and visual disturbances developed, including visual hallucinations and double vision that predominantly occurred on the right side, as well as visual

auras. Treatment with lacosamide was started for possible seizures, and visual symptoms abated.

After discharge from the other hospital, the patient returned home and attended a follow-up visit in the neurology clinic affiliated with this hospital. Treatment with lacosamide was continued, and in consultation with the neurosurgery team, plans were made for repeat cerebral angiography to be performed in 6 months.

One month later, and 5 months before the current presentation, generalized tonic-clonic seizure activity recurred. The patient was admitted to this hospital.

Dr. Zhu: CT of the head showed hyperattenuating embolic material with a linear branching pattern along the falx cerebri and cortical vessels surrounding the posterior superior sagittal sinus in the left cerebral hemisphere, findings suggestive of previous embolization of a vascular malformation (Fig. 2A). The area where the presumed subarachnoid hemorrhage had been noted previously was stable on both CT and MRI (Fig. 2B and 2C).

Dr. Amurthur: Cerebral angiography identified evidence of previous embolization but showed no evidence of a persistent shunt or venous thrombosis. The patient reported persistent visual disturbances, which did not have a correlate on

electroencephalography (EEG) and were suspected to be due to the presence of blood in the visual cortex.

Dr. Zhu: MRI of the head was also performed and revealed several foci of diffusion restriction in the right parietal lobe on diffusion-weighted sequences that corresponded to hyperintensity on FLAIR sequences — findings compatible with acute or early subacute ischemic infarcts, which were thought to be embolic (Fig. 2D, 2E, and 2F). Additional, nonspecific hyperintensities in the periventricular white matter were unchanged from the findings seen on MRI 18 months earlier.

Dr. Amurthur: The dose of lacosamide was in-

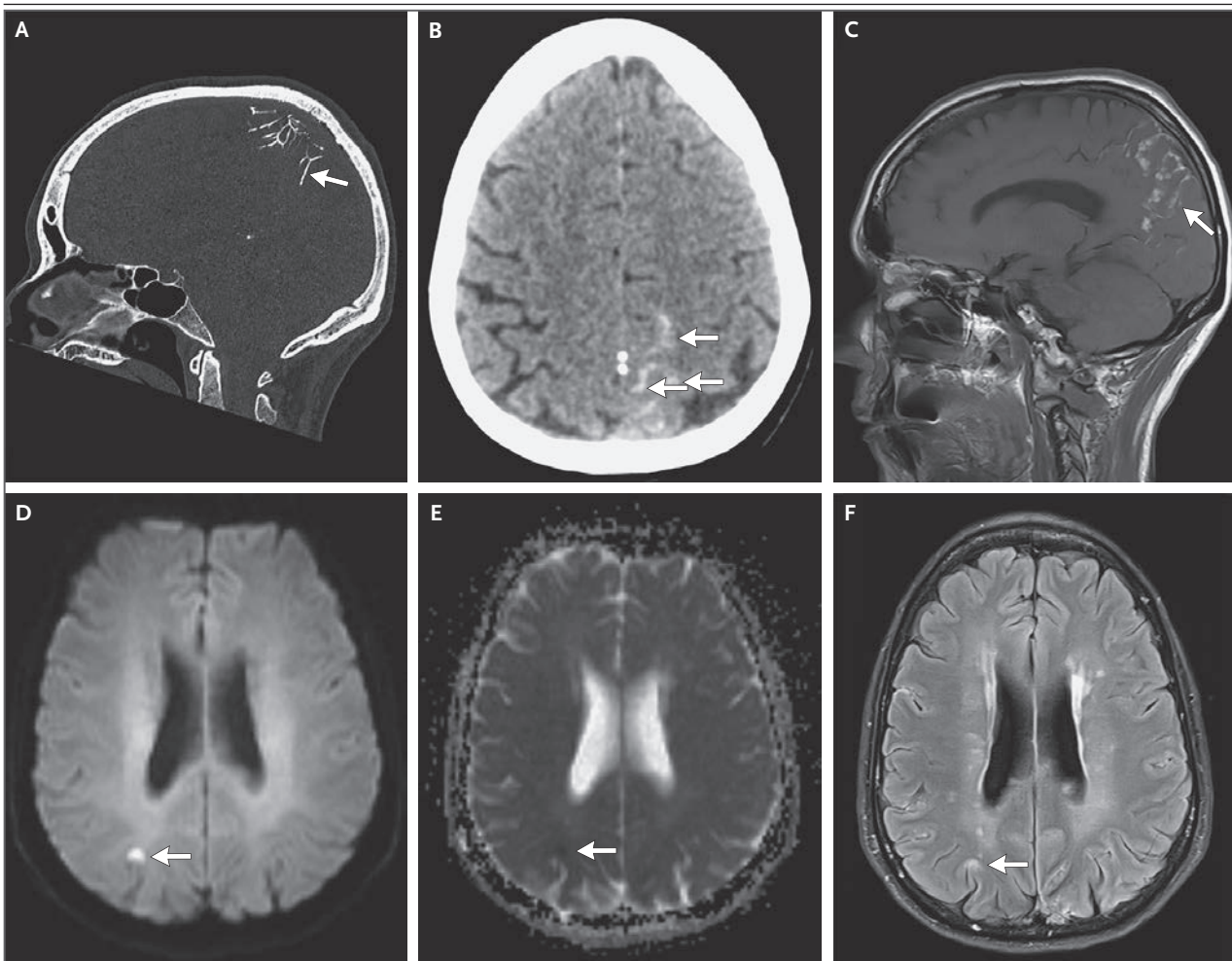


Figure 2. Imaging Studies of the Head, 5 Months before the Current Presentation.

A sagittal CT image (Panel A) shows hyperattenuating embolic material with a linear branching pattern along the falx cerebri (arrow). An axial CT image (Panel B) and a sagittal T1-weighted MRI (Panel C) show stability of the presumed subarachnoid hemorrhage (arrows). A punctate focus of hyperintensity in the right parietal lobe on an axial diffusion-weighted MRI (Panel D, arrow) corresponds to hypointensity on an axial apparent-diffusion-coefficient MRI (Panel E, arrow) and to hyperintensity on an axial fluid-attenuated inversion recovery MRI (Panel F, arrow).

creased, and the patient was discharged from the hospital.

One month later, and 4 months before the current presentation, the patient sustained a fall during an episode of dizziness and lightheadedness, and she was again admitted to this hospital. She had persistent visual disturbances involving the right visual field that she described as “flashing lights.”

Dr. Zhu: CT of the head again showed hyperdensity in the sulci of the posterior paracentral lobule in the left parietal lobe and, to a lesser degree, along the precentral sulcus of the right frontal lobe. No radiologically significant change from the previous study was noted.

Dr. Amurthur: Long-term EEG monitoring showed no evidence of seizure activity or an EEG correlate that would explain the visual disturbances. Examination by an ophthalmology consultant was normal. The patient’s presenting symptoms were thought to be due to presyncope, and no changes were made to the antiepileptic regimen.

One month later, and 3 months before the current presentation, blurry vision in both eyes and hoarseness developed. The patient had another generalized tonic-clonic seizure, and she was again admitted to this hospital.

Dr. Zhu: CT of the head showed a stable appearance of the hyperattenuation involving the sulci of the left parietal lobe, as well as a few scattered foci along the sulci of the right frontal lobe. These findings were thought to be consistent with previous embolization of the fistula. MRI of the head was also performed, and T1-weighted sequences showed nodular intrinsic hyperintensities in the sulci and leptomeninges of the left parietal and occipital lobes, with associated susceptibility signals that were suggestive of mineralization. These findings were similar to the MRI findings seen 2 months earlier. New nodular intrinsic hyperintensities were noted along the dorsal midbrain and in the posterior right sylvian fissure, the operculum in the right frontal lobe, the convexity of the right anterior temporal lobe, and the convexity of the left and right medial frontal lobes.

Dr. Amurthur: Lumbar puncture was performed, and cerebrospinal fluid (CSF) analysis showed 1 red cell per microliter, 1 nucleated cell per microliter, and normal levels of glucose and protein. Positron-emission tomography of the head showed low fluorodeoxyglucose uptake in the region where the left parieto-occipital sulcal abnormalities had

been seen on recent MRI, which indicated hypometabolism in that area. Treatment with levetiracetam was added, and the patient was discharged from the hospital. Cytologic evaluation of the CSF showed no malignant cells, and flow cytometry of the CSF showed no evidence of a monoclonal B-cell population.

One month later, and 2 months before the current presentation, the patient was evaluated in the otolaryngology clinic affiliated with this hospital because of hoarseness. Flexible laryngoscopy revealed immobility of the right vocal cord. The patient was also evaluated in the neurology clinic affiliated with this hospital because of increased sleepiness and episodes of tremulousness in the right arm and leg. Given the possibility of medication side effects, the course of levetiracetam was tapered off, and treatment with clobazam was started. The episodes of tremulousness became less frequent but did not abate, and the visual disturbances involving the right visual field persisted.

On the day of the current presentation, the episodes of tremulousness in the right arm and leg increased in frequency and length, lasting up to 10 minutes at a time. No loss of consciousness occurred. The patient presented to the emergency department of this hospital for evaluation.

The patient reported increasingly frequent visual disturbances involving the right visual field, which she described as “seeing rainbows, circles, and flashes.” Additional medical history included allergic rhinitis, vitamin D deficiency, and migraines. Medications included clobazam and lacosamide. She had no known allergies to medications. Before this illness, the patient had worked in real estate. She lived with her husband and two children in Massachusetts. She was a lifelong nonsmoker, consumed alcohol only occasionally, and did not use illicit drugs. Her family history included migraines in her father and brother, stroke in her mother, and melanoma in her father.

On examination, the temporal temperature was 36.5°C, the blood pressure 104/62 mm Hg, the heart rate 57 beats per minute, the respiratory rate 18 breaths per minute, and the oxygen saturation 96% while the patient was breathing ambient air. The body-mass index (the weight in kilograms divided by the square of the height in meters) was 18.5. The patient was alert and oriented. Language was fluent, and comprehension, repetition, and naming were intact. On examination of the cranial

nerves, saccadic breakdown was noted, along with slight vertical diplopia on left lateral gaze. Strength, tone, and muscle bulk were normal. Finger-to-nose testing showed a slight intention tremor and mild dysmetria that was greater on the left side than on the right side. The remainder of the neurologic examination was normal.

The blood levels of electrolytes, glucose, aspartate aminotransferase, alanine aminotransferase, alkaline phosphatase, bilirubin, albumin, and total protein were normal, as were the results of kidney-function tests. The complete blood count with differential count was normal.

A diagnostic test was performed.

DIFFERENTIAL DIAGNOSIS

Dr. Nancy Wang: I participated in the care of this patient and am aware of the final diagnosis. This 57-year-old woman had received a diagnosis of subarachnoid hemorrhage at another hospital 6 months before the current presentation and had undergone embolization of a suspected pial arteriovenous fistula. However, her symptoms continued to evolve and included recurrent subacute seizures that were escalating in intensity, as well as deficits in cranial-nerve function. After localizing the patient's symptoms within the nervous system, I will construct the differential diagnosis for her presentation, considering the temporal course of her illness and the findings on CSF analysis and neuroimaging.

NEUROANATOMICAL LOCALIZATION

Two years before the current presentation, the patient had frontotemporal and cervico-occipital headaches that lasted for several hours and were associated with photophobia. MRI showed non-specific findings, and she received a diagnosis of migraines. The median age at the onset of migraines is 23 to 25 years of age, and the prevalence is only 5% among women older than 50 years of age.¹ However, given the patient's family history of migraines and her clinical stability until 6 months before the current presentation, I suspect that these headaches were migraines that were unrelated to her current presentation.

Six months before the current presentation, a severe headache developed, although we are not told the location of the headache, whether it was similar to the previous migraines, or whether any features associated with increased intracranial

pressure were present. At that time, the patient also had visual auras, although we are not given details about the auras. Up to one third of patients with migraines have auras.¹ However, this patient's imaging of the head showed an abnormality in the left parietal lobe that was considered to be a more likely cause of her headache and could also be the cause of hallucinations in the lower quadrant of the right visual field.

The later development of shaking in the right arm and leg was indicative of focal motor seizures, which localize to the motor cortex of the left frontal lobe. Paralysis of the right vocal cord indicates possible dysfunction of the right vagus nerve. Vertical diplopia on left lateral gaze suggests partial palsy of the right trochlear nerve. Saccadic breakdown and dysmetria that is greater on the left side are both findings indicative of cerebellar dysfunction. In the absence of other motor and sensory deficits and of systemic abnormalities on imaging, the cerebellar signs and the deficits in vagus and trochlear nerve function localize the disease process to the subarachnoid space in the posterior fossa and skull base. In summary, the progression of this patient's symptoms — from those indicating localization to the supratentorial and cortical structures to those indicating localization to the posterior fossa and skull base — suggests a diffuse process affecting the subarachnoid space and leptomeninges.

SUBARACHNOID HEMORRHAGE

A diagnosis of subarachnoid hemorrhage was initially made on the basis of findings on CT of the head performed at another hospital after the patient had presented with severe headache, confusion, and generalized tonic-clonic seizure activity. Thunderclap headache, which is classically characterized as the "worst headache of the patient's life,"² is the most common presenting symptom of spontaneous (nontraumatic) subarachnoid hemorrhage. In up to half of patients with spontaneous subarachnoid hemorrhage, altered consciousness may occur,³ as well as seizures. Therefore, this patient's initial symptoms fit well with the presumed diagnosis.

Location is an important clue in determining the cause of subarachnoid hemorrhage. Non-traumatic convexal subarachnoid hemorrhage accounts for approximately 6% of all cases of subarachnoid hemorrhage.⁴ Reversible cerebral vasoconstriction syndrome is the most common

cause in patients younger than 60 years of age, often manifesting as thunderclap headache with or without focal neurologic symptoms due to ischemia or hemorrhage.⁵ Cerebral amyloid angiopathy is a common cause in patients older than 60 years of age, with the most common presenting symptoms being transient focal neurologic symptoms (also called “amyloid spells”).⁵ This patient was thought to have convexal subarachnoid hemorrhage that was caused by rupture of a fistula between the left middle meningeal artery and the superior sagittal sinus, and the fistula was embolized.

Neurologic complications of subarachnoid hemorrhage can include diffuse cerebral edema, hydrocephalus, seizures, and delayed cerebral ischemia. Delayed cerebral ischemia may be due to cortical spreading depolarization,⁶ which is characterized by large, depolarizing waves that propagate slowly across the cortex, with either transient hyperperfusion or spreading hypoperfusion causing ischemic injury. Delayed cerebral ischemia complicating subarachnoid hemorrhage could explain this patient’s transient visual disturbances without an EEG correlate. However, she had progression of clinical and imaging findings, which would be unexpected with subarachnoid hemorrhage after effective embolization of the source. Repeat angiography did not show a persistent shunt or venous thrombosis that would explain the progressive imaging findings. Although an inflammatory or granulomatous response to embolization material could be considered, the diffuse distribution of leptomeningeal changes would be unusual with this condition. I will therefore consider other potential diffuse processes involving the subarachnoid space and leptomeninges.

INFECTION

Could this patient have chronic infectious meningitis? She did not have meningeal signs, such as nuchal rigidity, but these signs are less common in patients with chronic infectious meningitis than in those with acute meningitis. The presence of headaches, lethargy, deficits in cranial-nerve function, and seizures is consistent with chronic infectious meningitis, but the absence of fever is atypical. Infectious causes of chronic meningitis include tuberculosis, Lyme disease, and syphilis, with cryptococcosis being the most common cause in immunocompromised patients.⁷ Coccidioidomycosis, histoplasmosis, and blastomycosis

should be considered in patients presenting in regions where the diseases are endemic, including the Southwest, Upper Midwest, and Mississippi River Valley regions of the United States. Chronic enteroviral meningitis can occur in patients with agammaglobulinemia or in those who are receiving B-cell-depleting immunotherapy.^{8,9} Patients who have received epidural injections should be evaluated for chronic fungal meningitis; an outbreak that occurred in 2012 was traced back to contaminated glucocorticoids.^{10,11} Chronic parasitic meningitis could be considered in the appropriate clinical context, but this patient did not have any pertinent epidemiologic risk factors. In addition, in immunocompetent patients with chronic infectious meningitis, CSF analysis almost always shows pleocytosis, an elevated protein level, and a low glucose level, and none of these findings were present in this patient.⁷

INFLAMMATORY OR AUTOIMMUNE DISEASE

Noninfectious inflammatory or autoimmune causes of chronic meningitis include sarcoidosis, rheumatoid arthritis, IgG4-related disease, Sjögren’s syndrome, and Still’s disease. The absence of systemic symptoms during the entire course of this patient’s illness makes all these disorders unlikely, although chronic meningitis may be the initial presenting symptom in patients with neurosarcoidosis or Sjögren’s syndrome.^{12,13} Other uveomeningeal syndromes — such as granulomatosis with polyangiitis, Behçet’s disease, and Vogt-Koyanagi-Harada syndrome — should be considered, particularly given this patient’s vision changes. However, she did not have evidence of uveitis on examination, and her symptoms were more suggestive of cortical dysfunction.

CANCER

Cancer cells that spread to the leptomeninges can have a nodular phenotype (with cells adherent to the pia mater) or a free-floating phenotype (with cells circulating in the CSF).¹⁴ The entire central nervous system (CNS) can be affected, and thus the range of neurologic symptoms is broad and nonspecific, spanning from cranial- and spinal-nerve dysfunction to headaches and mental-status changes. Impaired CSF resorption resulting from leptomeningeal metastases may lead to increased intracranial pressure, with symptoms including vision changes, positional headaches, nausea, and vomiting. However, progressive metastatic lepto-

meningeal disease often develops in the late stages of systemic cancer, and the degree of clinical deterioration seen in this patient would be unusual in the absence of active systemic disease.

Rare primary tumors that involve the leptomeninges include lymphoma, diffuse leptomeningeal glioneuronal tumor, and primary melanocytic tumors. Patients with primary leptomeningeal lymphoma typically present with multifocal symptoms and an abnormal CSF profile, specifically abnormal results on cytologic evaluation (in 67%) and on flow cytometry (in 80%).¹⁵ Patients with diffuse leptomeningeal glioneuronal tumor, a rare entity that was first described in the 2016 *World Health Organization Classification of Tumors of the Central Nervous System*, can have a more indolent course, and the disease may be localized to the spinal column. The presence of the *KIAA1549–BRAF* fusion is common, and the diagnosis often requires surgical biopsy.¹⁶ Primary melanocytic tumors, which arise from melanocytes in the leptomeninges that originate from the neural crest, include diffuse processes (meningeal melanocytosis and meningeal melanomatosis), as well as more circumscribed lesions (primary leptomeningeal melanoma and melanocytomas).¹⁷ Among the diffuse neoplasms, melanocytosis is typically benign and more common among children, whereas melanomatosis is associated with an aggressive course.

CSF ANALYSIS

Does this patient's normal CSF profile rule out cancer? Among patients with known leptomeningeal metastasis, the CSF cell count may be normal in 20 to 60%, the CSF protein level in approximately 20%, and the CSF glucose level in 40 to 60%.^{18–20} CSF cytologic evaluation is positive in more than 90% of patients with leptomeningeal metastasis, but this degree of sensitivity can be reached only with the performance of three high-volume lumbar punctures.^{21,22} In patients with hematologic cancer, the addition of CSF flow cytometry leads to higher sensitivity than the use of cytologic analysis alone.²³ Recent advances in molecular testing have improved the ability to diagnose cancers on the basis of circulating tumor cells and circulating tumor DNA, but this particular testing was not performed in this patient.^{24–26}

INTRINSIC HYPERINTENSITIES

A key imaging finding in this patient was intrinsic hyperintensities on T1-weighted MRI. This

feature narrows the differential diagnosis because most of the previously mentioned entities are associated with gadolinium-enhancing hyperintensity on T1-weighted imaging. Substances that are intrinsically hyperintense on T1-weighted MRI include hemoglobin-degradation products, lipid-containing lesions (e.g., dermoid cysts), highly proteinaceous lesions (e.g., craniopharyngioma), melanin, and mineralization products (e.g., calcifications).²⁷ This patient had no evidence of a ruptured cyst, and evolution over a period of 6 months would not be expected with subarachnoid hemorrhage. Therefore, I thought that a primary melanocytic tumor was the most likely explanation for this patient's presentation and clinical course, and I recommended meningeal biopsy as the next step.

DR. NANCY WANG'S DIAGNOSIS

Primary melanocytic tumor.

DIAGNOSTIC TESTING

Dr. Zhu: MRI of the head was performed at the time of the current presentation. T1-weighted sequences showed an increase in the intensity and distribution of the hyperintensities in the left parietal and occipital lobes and progression of the leptomeningeal thickening and sulcal nodularity throughout the cerebral hemispheres (Fig. 3A and 3B) as compared with the findings seen 3 months earlier (Fig. 3D and 3E). There was also enlargement of the ventricles (Fig. 3C) as compared with the findings seen 3 months earlier (Fig. 3F), which is suggestive of developing hydrocephalus.

Dr. Aman B. Patel: Given the possibility of cancer, right frontal craniotomy with brain biopsy was performed. The dura appeared abnormal, and black tissue was noted. The pia mater was opened, and multiple samples of the abnormal tissue were obtained for evaluation.

Dr. Anthony R. Russo: Intraoperative smear and frozen-section examinations revealed diffuse sheets of round cells with prominent nucleoli and dark cytoplasmic pigment, features consistent with a melanocytic neoplasm (Fig. 4A). Permanent histologic sections of the leptomeninges and frontal cortex showed a diffuse proliferation of malignant melanocytes in the subarachnoid space, invading cortical gray matter (Fig. 4B and 4C) and perivascular Virchow–Robin spaces (Fig. 4D). On immu-

nohistochemical staining, the malignant cells were positive for melanocyte-inducing transcription factor (Fig. 4E), a finding that confirms a melanocytic origin. Molecular testing revealed a p.R183Q variant in *GNAQ* (the gene encoding the guanine nucleotide-binding protein α q subunit). This mutation is often seen in primary CNS and uveal melanocytic tumors, distinguishing them from cutaneous melanomas, which are most commonly associated with *BRAF* mutations.¹⁷ Given the diffuse nature of the lesion seen on imaging, the invasion of the cortical gray matter seen on

histologic examination, and the characteristic findings seen on molecular testing, the lesion was diagnosed as diffuse meningeal melanomatosis.

PATHOLOGICAL DIAGNOSIS

Diffuse meningeal melanomatosis.

DISCUSSION OF MANAGEMENT

Dr. Patel: During the patient's hospitalization, CT of the chest, abdomen, and pelvis was performed

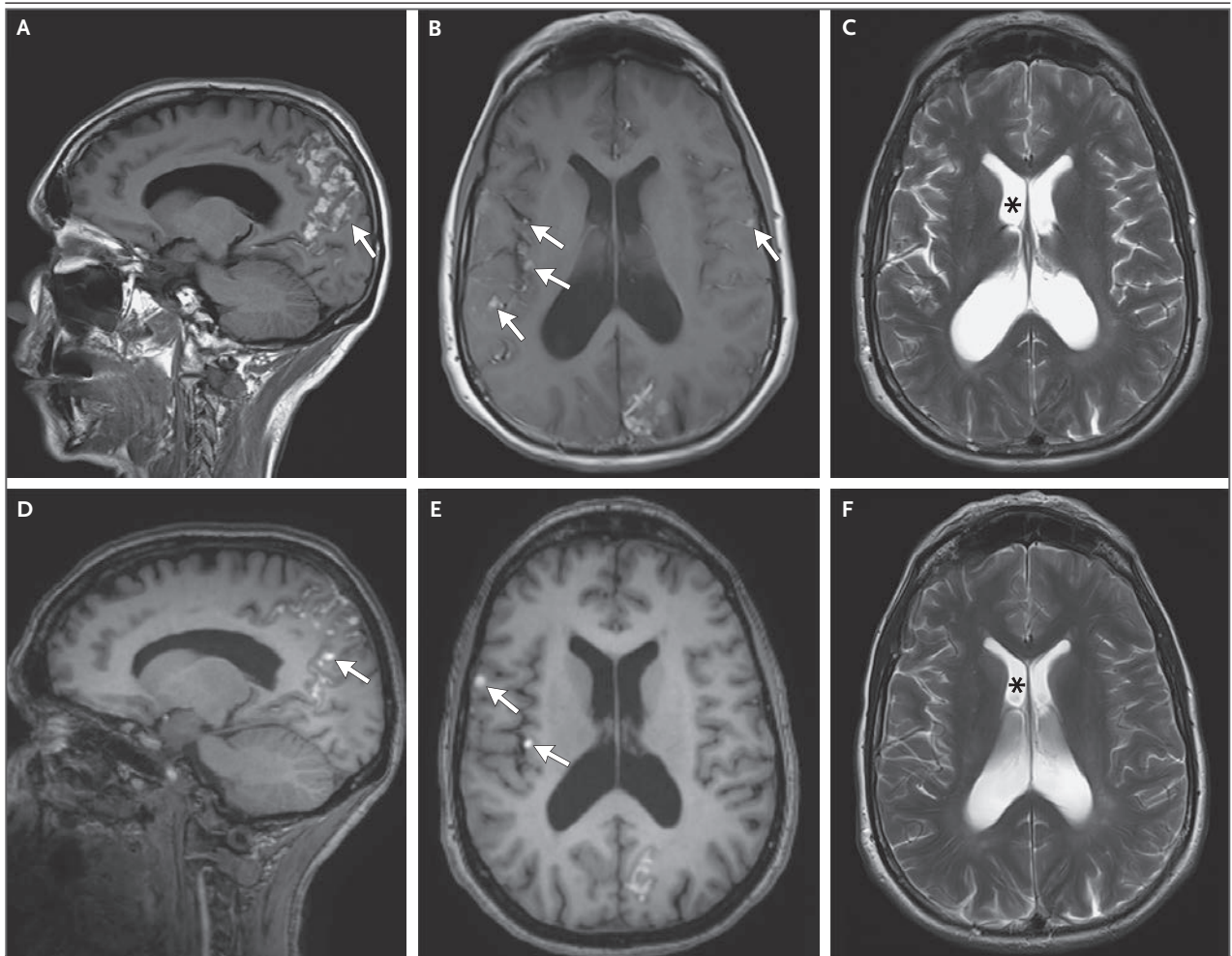


Figure 3. Current Imaging Studies of the Head.

Imaging studies obtained at the time of the current presentation (Panels A, B, and C) were compared with those obtained 3 months before the current presentation (Panels D, E, and F). Sagittal and axial T1-weighted MRIs obtained on the current presentation (Panels A and B, respectively) show an increase in the intensity and distribution of the hyperintensities in the left parietal lobe and additional leptomeningeal foci in the left and right cerebral hemispheres, as compared with the findings seen on sagittal and axial T1-weighted MRIs obtained 3 months earlier (Panels D and E, respectively; arrows). An axial T2-weighted MRI obtained on the current presentation (Panel C) shows enlargement of the lateral ventricles (asterisk), as compared with the findings seen on an axial T2-weighted MRI obtained 3 months earlier (Panel F, asterisk).

at the request of the neuro-oncology team and was negative for metastatic disease. However, MRI of the spine revealed multiple enhancing leptomeningeal lesions that were suggestive of

a metastatic process. Given the presence of minimally symptomatic hydrocephalus, elective placement of a ventriculoperitoneal shunt was planned. Approximately 2 weeks later, the patient un-

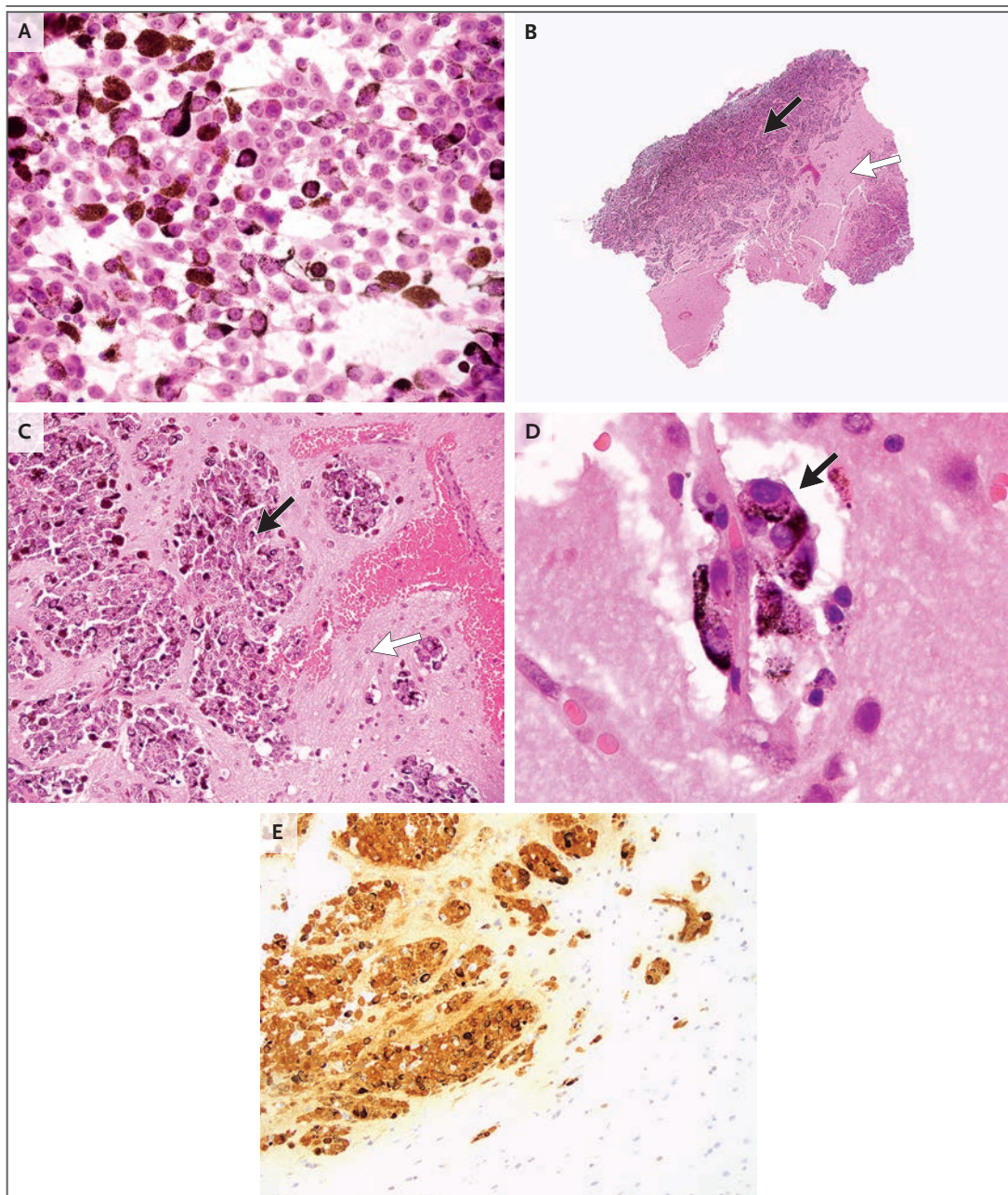


Figure 4. Biopsy Specimens of the Brain.

Hematoxylin and eosin staining of an intraoperative-smear specimen shows neoplastic melanocytes (Panel A). Hematoxylin and eosin staining of a cortical-biopsy specimen shows a proliferation of malignant melanocytes (Panels B and C, black arrows) invading cortical gray matter (Panels B and C, white arrows) and perivascular Virchow–Robin spaces (Panel D, arrow). Immunohistochemical staining for melanocyte-inducing transcription factor is positive (Panel E).

derwent placement of a ventriculoperitoneal shunt with a fixed pressure valve in the right frontal lobe, without complications. After the procedure, she continued to take her antiepileptic medications at home, and follow-up EEG showed no evidence of breakthrough seizures.

Dr. Donald P. Lawrence: Treatment for melanocytic tumors of the leptomeninges depends on the degree of tumor aggressiveness, which varies, and whether the tumor is localized or diffuse. Nonaggressive, localized tumors can be managed with surgery and radiotherapy, which result in durable disease control in some cases. Surgery and radiotherapy have a limited role in the management of aggressive, diffuse tumors.²⁸ These tumors rarely harbor *BRAF* mutations and are not amenable to treatment with the tyrosine kinase inhibitors that are used for cutaneous melanoma.²⁹

Given the lack of data on systemic therapy for meningeal melanoma, we extrapolated from data on such treatment for uveal melanoma, a disease that is also characterized by mutations in genes encoding G proteins that are not directly targetable.²⁹ Both uveal and meningeal melanoma have low tumor mutational burdens, and uveal disease responds only occasionally to treatment with immune checkpoint inhibitors.³⁰ We considered treatment with the bispecific fusion protein tebentafusp, which has been shown to recruit T cells and prolong survival in patients with metastatic uveal melanoma.³¹ However, the target antigen (gp100)

is presented in the context of HLA-A*02:01, and the patient did not have that HLA allele.

With the understanding that the likelihood of benefit was low, the patient opted for treatment with the immune checkpoint inhibitors ipilimumab and nivolumab. After the fourth dose, her cognitive function worsened, and headaches and nausea developed. She received supportive treatment. After she was found to have progressive disease on imaging, the patient elected to receive hospice care. The patient died approximately 5 months after the initial diagnosis.

FINAL DIAGNOSIS

Diffuse meningeal melanomatosis.

This case was presented at Neurology Grand Rounds.

Disclosure forms provided by the authors are available with the full text of this article at NEJM.org.

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